



Activity based
Project Report on
Artificial Intelligence
Project – I
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PROJECT REPORT: PHASE 1

TITLE: A* ALGORITHM FOR RESOURCE OPTIMIZATION IN LOGISTICS IN TERMS OF RESOURCE UTILIZATION MAXIMIZATION

1. INTRODUCTION

The fundamental crisis in modern logistics is the "Siloed Optimization" of transport assets. Most existing logistical frameworks utilize a decoupled software architecture: the packing of a vehicle is treated as a 3D Bin Packing Problem (3DBPP) solved statically at the warehouse, while the routing is treated as a Traveling Salesman Problem (TSP) solved by GPS-style algorithms. Once a vehicle departs the loading bay, its internal capacity is essentially "locked" until it reaches its final destination, regardless of its load efficiency.

Formally, this is a failure to solve the **Capacitated Vehicle Routing Problem with 3D Loading Constraints (3L-CVRP)** in a dynamic environment. This lack of real-time spatial-route synchronization leads to massive resource waste. Industry statistics indicate that nearly 25% of commercial trucks on roads are traveling completely empty, and over 50% are traveling with less than half of their volumetric capacity utilized.

Furthermore, the "Last-Mile" delivery segment exacerbates this inefficiency. Small, frequent deliveries often force large-capacity vehicles to enter high-congestion urban zones while mostly empty. Without a mathematical mechanism to facilitate mid-route "fusions"—where a large, well-utilized truck can absorb the cargo of a smaller, under-utilized agent—the industry remains trapped in a cycle of high operational overhead. The challenge for AI is to identify "Convergence Nodes" in a vast, dynamic road network where these fusions can occur without violating strict delivery time-windows.]

2. OBJECTIVE

The primary objective of this project is to improve logistics efficiency by optimizing delivery routes and maximizing vehicle capacity utilization using a modified heuristic search approach based on the **A*** algorithm.

Traditional route planning systems primarily focus on finding the **shortest path between locations**, but they do not consider the **dynamic utilization of logistics resources**, such as vehicle capacity and mid-route collaboration between vehicles. This often leads to inefficient delivery operations where trucks travel partially empty, resulting in increased fuel consumption, operational costs, and unnecessary vehicle deployment.

The objectives of this project are as follows:

I. Implement A* for Route Optimization

The first objective is to implement the **A*** pathfinding algorithm to determine efficient routes within a logistics network modeled as a directed weighted graph. The algorithm evaluates possible paths using both the actual travel cost and an estimated cost to the destination, allowing it to efficiently compute optimal routes for delivery vehicles.

ii. Extend A* to a Multi-Agent Logistics Environment

Another objective is to adapt the traditional **A*** search algorithm to operate in a **multi-agent environment**, where multiple delivery vehicles independently compute routes while operating within the same road network. This allows the system to simulate real-world logistics operations involving multiple vehicles simultaneously.

iii. Introduce Fusion Nodes for Load Consolidation

The project aims to introduce **fusion nodes**, which represent strategic junctions where vehicles can potentially meet and consolidate their loads. By identifying intersections in vehicle routes, the system evaluates opportunities for combining deliveries into fewer vehicles.

iv. Maximize Vehicle Capacity Utilization

A key objective is to maximize the **volumetric utilization of vehicles**. When two vehicles meet at a fusion node, the algorithm checks whether their combined cargo can fit within the capacity of a single vehicle. If feasible, the cargo is transferred and the redundant vehicle is removed from the delivery process.

v. Modify the A* Heuristic with Fusion Potential

To support route collaboration, the project introduces a modified heuristic component in the **A*** evaluation function that considers the **fusion potential of nodes**. This encourages the algorithm to prefer routes where vehicles have higher chances of meeting and merging loads, thereby improving overall resource utilization.

vi. Reduce Empty Miles in Logistics Operations

Another objective is to reduce the occurrence of **empty or underutilized vehicle travel**, commonly known as “empty miles.” By consolidating deliveries during transit, the system aims to minimize the number of vehicles required to complete a set of delivery tasks.

vii. Demonstrate a Simulation-Based Logistics Optimization Model

Finally, the project aims to develop a **simulation framework** that demonstrates how heuristic search algorithms like **A*** can be extended beyond simple pathfinding to support collaborative decision-making in logistics networks.

3. INTRODUCTION TO AI IN LOGISTICS

The application of Artificial Intelligence in logistics has transitioned from basic automation to complex, multi-objective optimization. At its core, logistics routing is a highly constrained variation of the NP-hard Vehicle Routing Problem (VRP). Traditionally solved using basic heuristics or meta-heuristics (like Genetic Algorithms or Simulated Annealing), the modern requirement for "Resource Maximization" demands an algorithm capable of navigating high-dimensional state spaces where constraints are geographical, volumetric, and temporal.

In the context of this research, resource utilization is mathematically defined as the ratio of a vehicle's occupied capacity—accounting for both 3D volumetric displacement (m^3) and gravitational weight (

kg)—to its total potential capacity. As global supply chains face increasing regulatory pressure to reduce carbon footprints (Green Logistics), the "linear delivery model" is proving economically and environmentally obsolete.

This project seeks to leverage the informed search capabilities of the A* algorithm to create a "non-linear" routing system. By implementing a modified heuristic that treats empty truck space as a quantifiable "cost" and consolidated loads as a "search reward," we simulate a self-organizing logistics network. This approach transforms the A* algorithm from a simple pathfinder into a sophisticated resource manager, where the paths taken are dictated by the probability of resource fusion and fleet consolidation, ultimately driving down operational expenditure (OPEX) and emissions.

4. COMPREHENSIVE OVERVIEW OF THE PROBLEM

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Table 1: Comparative Analysis of Traditional vs. Intelligent Logistics Frameworks
5. ROLE OF THE A* ALGORITHM IN MULTI-OBJECTIVE OPTIMIZATION

The A* algorithm is uniquely suited for logistics optimization due to its properties of **Optimality** and **Completeness** (provided the heuristic is admissible). In a search space as dense as a city's road network, A* utilizes a heuristic $h(n)$ to prune the search tree, ensuring the algorithm does not waste computational resources expanding nodes that lead away from the goal.

While Deep Reinforcement Learning (DRL) is popular, it suffers from "black-box" unpredictability and high training costs. A*, conversely, provides deterministic, explainable AI routing

Dimension	Traditional Pathfinding	Optimized Resource Logistics	AI-Driven Impact

	(Status Quo)	(Proposed)	
Agent Autonomy	Isolated; Agents act as independent nodes.	Collaborative; Agents act as a swarm.	Elimination of redundant journeys.
State Definition	2D (x, y) or 3D (x, y, t)	High-Dimensional (x, y, t, Vol, W)	Full awareness of resource state.
Heuristic Goal	Distance/Time Minimization.	Utility/Resource Maximization.	Increased profitability per trip.
Fleet Dynamics	Fixed Fleet Size per shift.	Dynamic "Fleet Collapse" via Fusion.	Reduced traffic and carbon footprint.

, which is crucial for enterprise logistics. In this project, the role of A* is expanded from a single-agent search to a **Multi-Agent Coordination (MAC)** kernel. The algorithm projects the future states of multiple vehicles simultaneously onto a shared "Space-Time-Volume" graph.

The "Intelligence" of our implementation lies in the heuristic's ability to evaluate the **Opportunity Cost**. The algorithm must calculate whether a 10% increase in travel distance to facilitate a fusion is mathematically "cheaper" than the 100% fuel and labor cost of running two separate, under-utilized vehicles. This transition moves A* from a distance-calculator to a multi-variable optimizer, allowing it to solve for both route efficiency and resource density concurrently.

6. LITERATURE REVIEW (2020–2026)

The following research papers form the academic foundation for this project, spanning the most recent advancements in heuristic search and logistics.

No.	Paper Title	Author/Year	Algorithm/Method	Key Findings
1	Multi-Agent Collaborative Path Planning Algorithm with Multiple Meeting Points	Applied Sciences (2024)	<i>Co-DPSIPP (Modified A)*</i>	Base Paper: Establishes the math for agents to synchronize at "Meeting Points" for task handovers.
2	Sustainable Route Planning for Urban Logistics	MDPI (2023)	Hybrid A* & GA	Optimized paths for fuel efficiency but

				ignored mid-route load merging.
3	Dynamic Resource Allocation in Industry 5.0	IEEE (2025 - Early Access)	Heuristic Search	Discusses "Resource Reciprocity" where vehicles trade tasks based on proximity.
4	Volumetric Constraints in Multi-Modal Logistics	Logistech (2022)	Greedy Best-First Search	Highlighted the failure of 2D algorithms in 3D cargo optimization.

Critical Analysis of Base Paper: The 2024 base paper (*Multi-Agent Collaborative Path Planning...*) provides the necessary algorithmic logic for identifying "Meeting Points" in a graph. However, it treats every agent as an abstract mathematical point with no physical mass or volume constraints. This project addresses this critical oversight by integrating **Volumetric Capacity** as the primary search constraint, ensuring that an algorithmic "Meeting" only triggers if a physical "Fusion" is dimensionally possible.

7. RESEARCH GAP: THE "RESOURCE-AWARENESS" VOID

Despite the maturity of pathfinding research, a critical void exists in the realm of "Integrated Resource Consolidation."

The Volumetric Neglect: Current AI models focus primarily on Time Windows (VRPTW) and distance constraints. They do not consider that the 3D volume of a truck changes dynamically at every delivery and pickup node. Because existing algorithms are "blind" to the actual space remaining in the truck during the search propagation, they cannot suggest opportunistic consolidations that lie marginally off the primary path.

Static Fleet Assumptions: Standard research models assume a rigid, static fleet constraint: if ten trucks are dispatched for a shift, those same ten must remain active until all routes are completed. There is a profound lack of research into **"Dynamic Fleet Collapse,"** where an AI identifies mid-journey that the remaining cargo of Truck A can fit into the empty space of Truck B, allowing Truck A to be "retired" early to conserve energy and labor.

Spatio-Temporal Synchronization Complexity: Existing collaborative models often force vehicles to wait for long periods at a meeting point, negating the efficiency gained. There is a gap in creating a heuristic that balances the "Wait Time Penalty" against the "Fusion Reward."

8. PROPOSED METHODOLOGY: VOLUMETRIC FUSION A* (VF-A*)

The proposed VF-A* algorithm operates on a high-dimensional directed acyclic graph (DAG) where each node is defined by its geographic coordinates, current timestamp, and the available volume/weight of the agent.

A. Mathematical Formulation of the Modified Heuristic

The core AI innovation is the introduction of the **Fusion Potential Coefficient** (Φ) into the standard A* evaluation function:

$$f(n) = g(n) + h(n) - \Phi(n)$$

- **$g(n)$** : The exact accumulated cost from the origin to node n (calculated via distance and fuel burn rate).
- **$h(n)$** : The admissible estimated cost to the target destination (e.g., Euclidean or Manhattan distance).
- **$\Phi(n)$** : The Fusion Potential. This acts as a "Negative Cost" or algorithmic search reward.

When the A* algorithm expands a node n , it scans the global registry for neighboring agents. If an Agent B is projected to intersect node n within an acceptable time window (Δt), the algorithm verifies the **Volumetric Constraint**:

$$(Volume_{Agent_A} + Volume_{Agent_B}) \leq Capacity_{Max_Vehicle}$$

If this

constraint is satisfied, the value of $\Phi(n)$ increases exponentially. This mathematically "pulls" the search tree towards node n , forcing the pathfinder to prioritize the consolidation event over a slightly shorter, solitary route.

B. Algorithmic Execution Logic

The project implementation follows a synchronized four-stage computational process:

- **Parallel Search Execution**: The central AI initiates individual A* searches for all active vehicle orders. Each agent builds its own search tree while continuously broadcasting its predicted future states and volumetric capacity to a global coordination registry.
- **Spatio-Temporal Intersection Detection**: The algorithm identifies "Conflict Nodes"—points in the road network where two vehicles can arrive at the same time without violating their respective customer delivery deadlines.
- **The Fusion Trigger (Handover Simulation)**: Once a fusion node is identified via the $\Phi(n)$ reward, the software simulates the transfer of virtual "Package Objects" from the "Giving Agent" to the "Receiving Agent," recalculating the new payload weight and volume.
- **Resource Recycling & Route Re-optimization**: The "Giving Agent" (now 100% empty) is instantly removed from the active delivery queue and assigned a fast-track return path to the

nearest depot. The "Receiving Agent" proceeds with a maximized resource load, effectively executing the work of two vehicles.

Table 2: Key Computational Components of the VF-A* Optimizer

Component	AI Responsibility	Search Logic Integration
State-Space Manager	Manages high-dimensional nodes.	Expands search beyond 2D (x, y) to (x, y, t, v, u) .
Fusion Evaluator	Calculates volumetric/time compatibility.	Triggers the Φ reward in the heuristic if valid.
Volumetric Tracker	Tracks 3D space occupied vs. total.	Updates the $g(n)$ cost based on current weight/load.
Path Collapser	Merges two active route objects.	Prunes active agents, reducing overall computational load.

9. SEARCH SPACE & DATASET

- Graph Representation (Search Space):

The logistics network is modeled as a directed graph

$$G = (V, E)$$

- $V \rightarrow$ logistics hubs (nodes)
- $E \rightarrow$ road connections between hubs (edges)

Each edge has attributes such as:

- distance
- traffic weight
- travel time

These values determine the **cost of moving between nodes** when the A* search algorithm evaluates possible routes.

- Node Types:

Nodes are divided into three categories:

- Pickup Nodes:** Warehouses or distribution centers where goods originate.
- Delivery Nodes:** Customer locations where goods must be delivered.
- Fusion Nodes:** Strategic intersections used for routing decisions and path merging.

This categorization helps the algorithm understand **where goods start, where they must go, and**

where routes can be optimized.

- **Resource Features:**

Each vehicle or delivery resource is described by:

- Current Load Volume L_c
- Maximum Payload L_{max}

Resource utilization is measured as:

$$U = \frac{L_c}{L_{max}}$$

The goal is to **maximize utilization while minimizing travel cost and idle capacity.**

- **Dataset Description:**

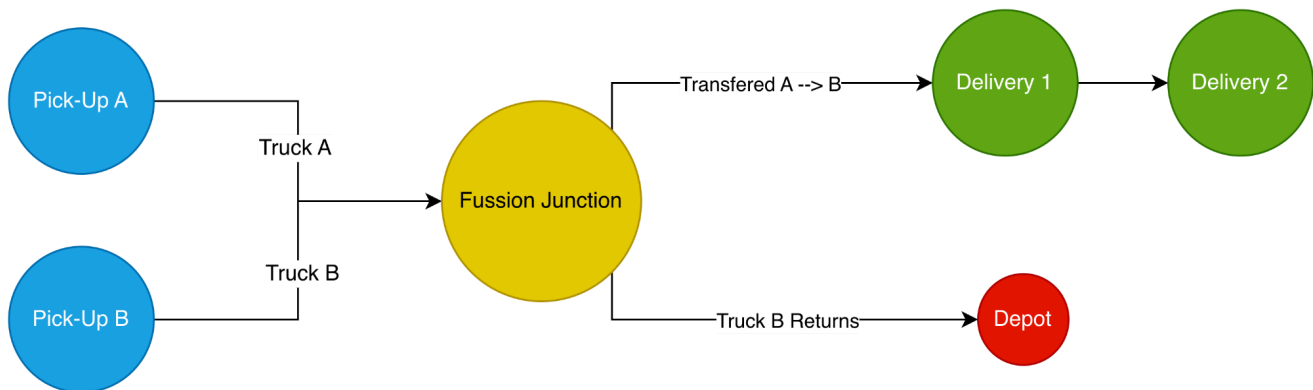
A **synthetic dataset** is used to simulate a logistics network based on the industrial corridor around:

- Hinjewadi
- Chakan
- Talegaon

Dataset characteristics:

- **50+** delivery nodes
- **10+** distribution centers
- road connections with distance and traffic weights

This dataset allows testing of routing strategies and resource allocation in a **realistic logistics environment.**



Dig 1. Search Space Graph

10. SYSTEM DESIGN & DIAGRAMS

A. Architecture Diagram (Description)

The proposed system follows a 3-Tier Architecture, which separates data management, processing logic, and visualization. This separation improves scalability, maintainability, and performance of the logistics optimization system.

- **Data Tier:**

The Data Tier stores all persistent logistics information required by the system.

Main components include:

- **Road Network Graph**
Represents the transportation network as nodes and edges, where nodes represent logistics hubs and edges represent road segments.
- **Package Registry**
Stores details of all packages such as weight, volume, and destination node.
- **Vehicle Registry**
Maintains information about available vehicles including maximum payload capacity and current load status.

This tier acts as the data source for the optimization engine.

- **Application Tier (VF-A Engine):**

The Application Tier contains the core optimization logic, referred to as the *VF-A Engine (Vehicle Fusion A Engine)*.

This component is responsible for:

- Running the A* search algorithm for route discovery
- Performing **multi-agent pathfinding** where multiple vehicles are optimized simultaneously
- Detecting **fusion points** where delivery routes can merge to improve resource utilization
- Minimizing travel cost while maximizing vehicle capacity usage

The engine processes data from the Data Tier and produces optimized route plans.

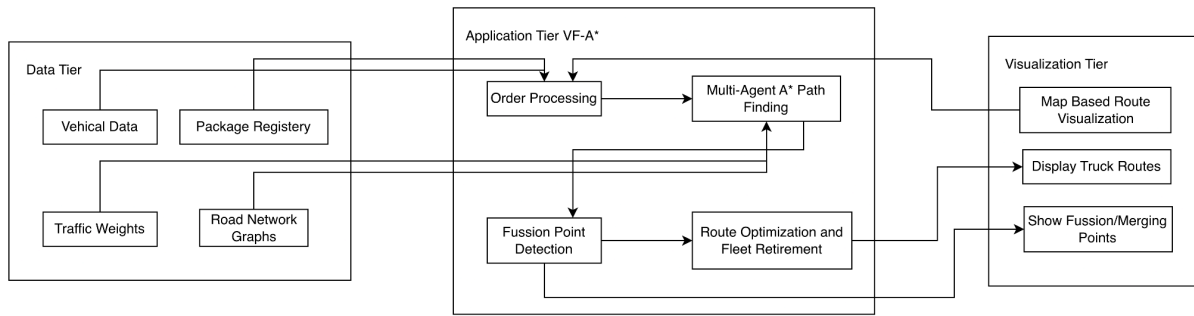
- **Visualization Tier:**

The Visualization Tier provides a map-based interface that displays the optimized routes generated by the system.

Key features include:

- Visualization of pickup and delivery nodes
- Display of vehicle paths across the logistics network
- Highlighting of **fusion points where routes merge**
- Real-time route updates

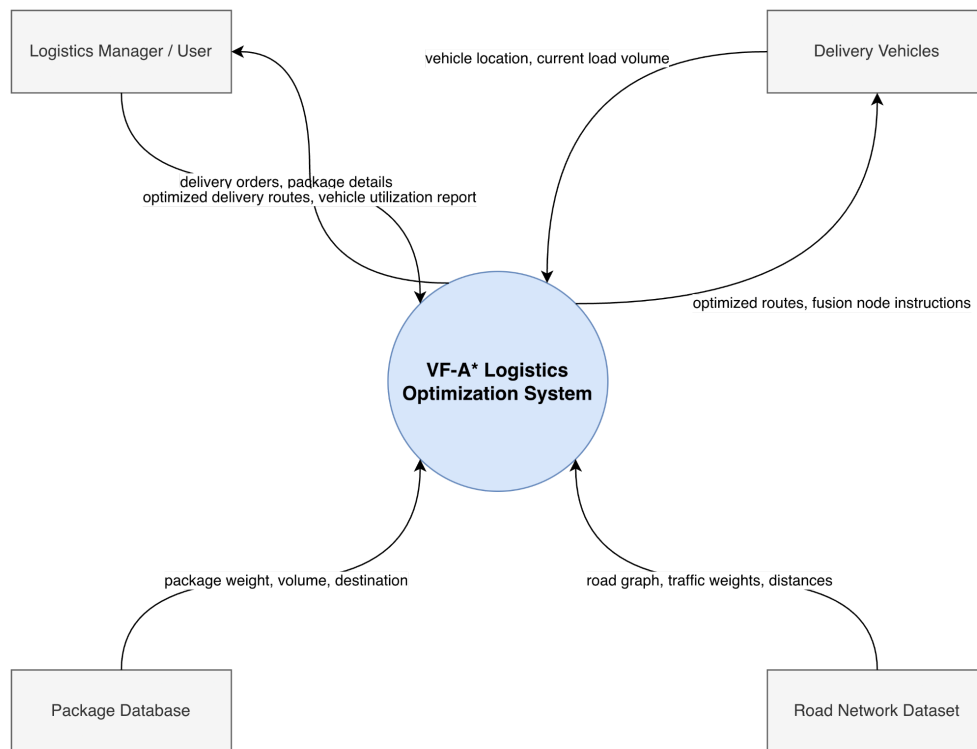
This layer allows operators to easily **monitor logistics flow and evaluate optimization performance**.



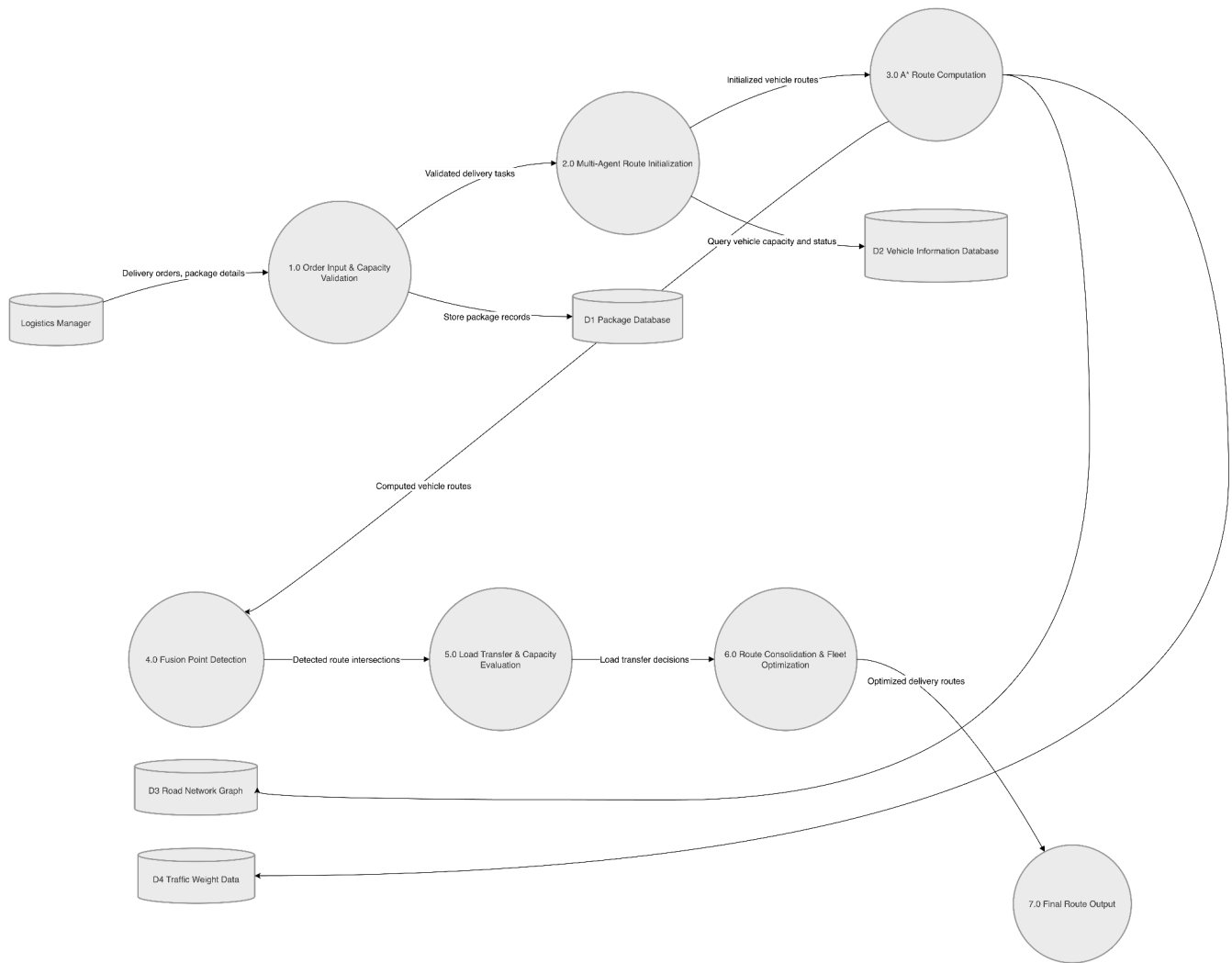
Dig 2. Architecture Diagram

B. Data Flow Diagram (DFD) - Level 1

- **Process 1.0:** Order Input & Capacity Validation.
- **Process 2.0:** Multi-Agent A* Search Initializer.
- **Process 3.0:** Fusion Point Detection & Logic.
- **Process 4.0:** Route Consolidation & Final Output.



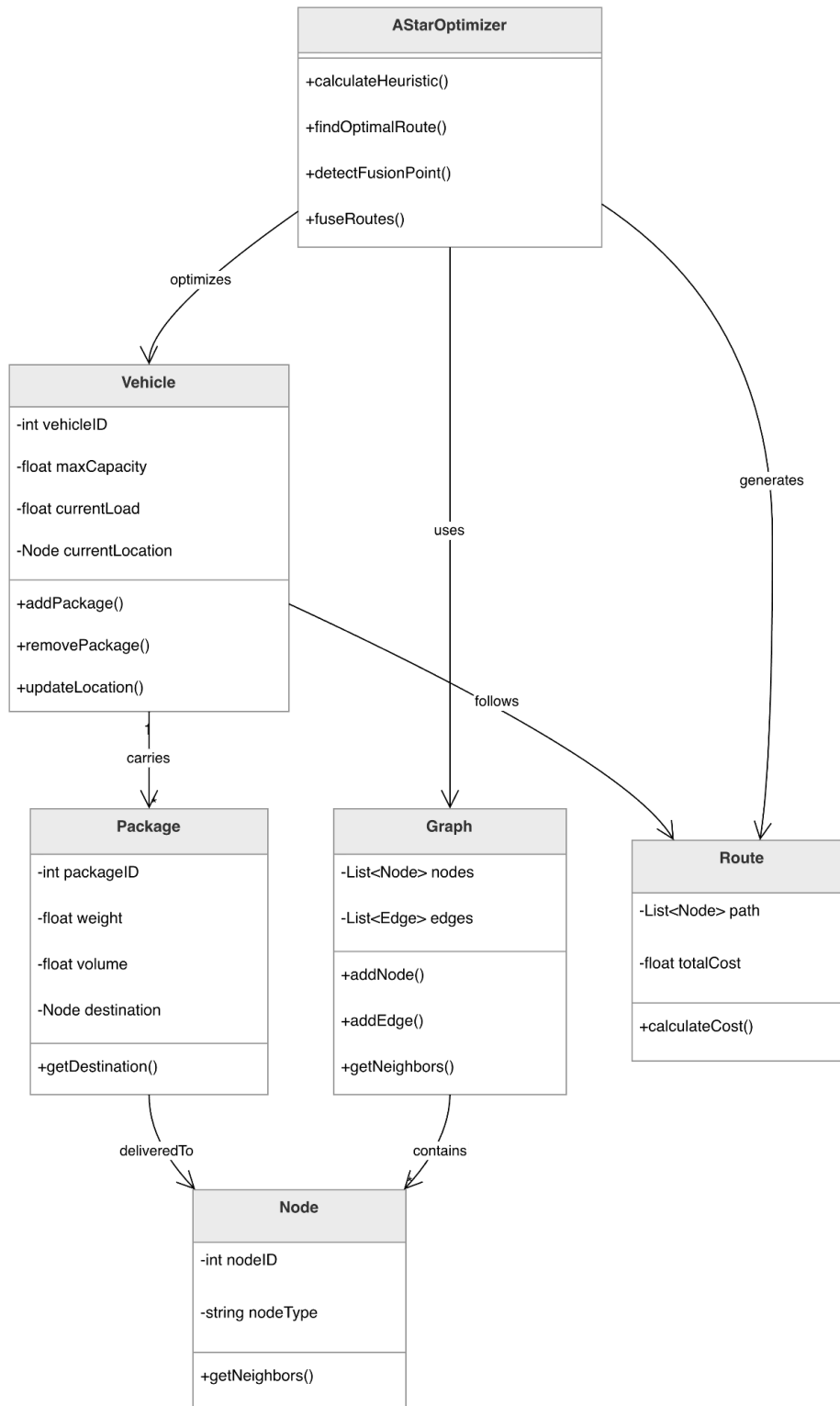
Dig 3.1. DFD Level 0



Dig 3.1 DFD Level 1

C. UML Class Diagram

- class Vehicle: Attributes (ID, MaxVol, CurrentVol, CurrentPos).
- class Package: Attributes (Weight, Volume, Destination).
- class AStarOptimizer: Methods (calculateHeuristic, findMeetingPoint, fuseRoutes).



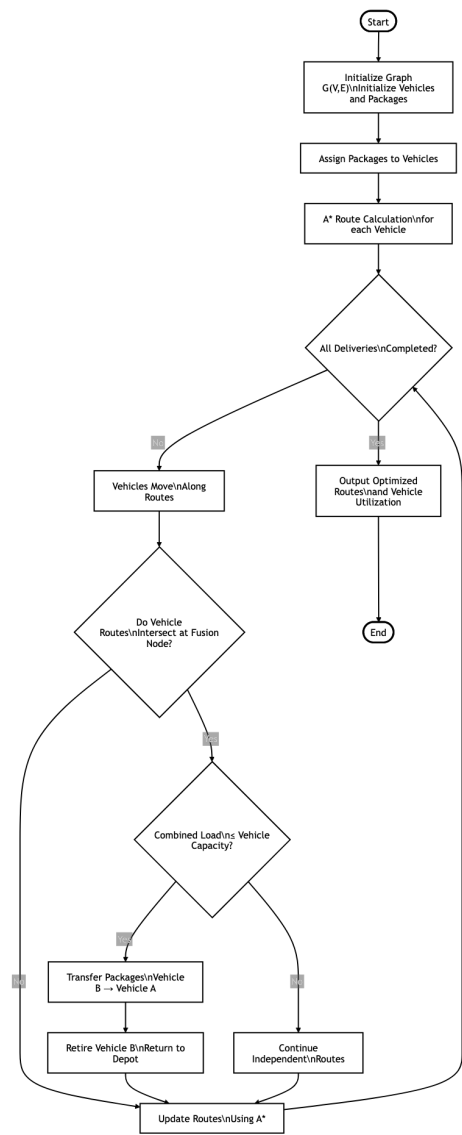
Dig 4. UML Diagram

11. TEST CASES

Test ID	Scenario	Expected Result
TC-01	Two 40% filled trucks cross	Algorithm fuses loads;

	paths at a junction.	Truck A becomes 80% full; Truck B is released.
TC-02	Total load of intersecting trucks exceeds 100%.	Fusion is rejected; trucks continue on individual shortest paths.
TC-03	1 Truck with 30% Capacity	Truck will continue its journey till destination without fusion

Flowchart:



Algorithm:

Step 1: Model the logistics network as a **directed weighted graph $G(V,E)$** where nodes represent pickup nodes, delivery nodes, and fusion nodes.

Step 2: Initialize all vehicles with their **current position**, **load volume**, and **maximum payload capacity**.

Step 3: Assign packages to vehicles based on their **pickup locations**.

Step 4: For each vehicle, compute an initial route to its destination using the **A* search algorithm**.

Step 5: Monitor vehicle routes to detect **intersections at fusion nodes.**

Step 6: When two vehicles reach the same fusion node, calculate their **combined load volume**.

Step 7: If the combined load is **less than or equal to the vehicle capacity**, transfer packages from one vehicle to the other.

Step 8: Update the receiving vehicle's load and mark the other vehicle as **inactive or return-to-base**.

Step 9: If the combined load exceeds capacity, allow both vehicles to **continue their original routes**.

Step 10: Update routes using A* if necessary and continue monitoring for further fusion opportunities.

Step 11: Repeat the process until **all packages are delivered**.

Step 12: Output the **optimized delivery routes and vehicle utilization results.**

Pseudo Code:

```

procedure VF_A_STAR(Graph G, Vehicles V, Packages P)

```

```
initialize vehicle states
```

assign packages to vehicles based on pickup locations

for each vehicle v in V

```
v.route ← A STAR SEARCH(G, v.startNode, v.destination)
```

while pending deliveries exist

for each vehicle v_1 in V for each vehicle v_2 in V

if $v_1 \neq v_2$

```
meetingNode ← FIND_ROUTE_INTERSECTION(v1.route, v2.route)
```

if meetingNode is not null AND meetingNode.type = FusionNode

```
combinedLoad ← v1.currentLoad + v2.currentLoad
```

```
if combinedLoad ≤ v1.maxCapacity
```

TRANSFER PACKAGES(from v2 to v1)

v1.currentLoad ← combinedLoad
mark v2 as inactive

v2.route ← A_STAR_SEARCH(G, meetingNode, Depot)

else

continue normal routing

for each active vehicle v

update v.position along its route

if new fusion opportunity detected

recompute route using A_STAR_SEARCH

return optimized routes

11. EXPECTED OUTCOME

- **Utilization Increase:**

The proposed system aims to achieve an **expected 20–25% increase in average volumetric utilization per trip**.

Volumetric utilization refers to how much of a vehicle's total cargo capacity is actually used during transportation.

$$Utilization = \frac{Current\ Load\ Volume}{Maximum\ Vehicle\ Capacity}$$

- **Resource Efficiency:**

Another expected outcome is a **reduction in the number of vehicles required to complete a fixed set of delivery orders**.

Traditional logistics systems often assign vehicles independently, which can result in:

- underutilized vehicles
- redundant travel routes
- increased operational cost

The proposed approach identifies opportunities to **merge delivery paths at fusion nodes**, allowing multiple deliveries to be handled by fewer vehicles. This leads to:

- reduced fuel consumption
- lower operational cost
- improved fleet utilization

Overall, the system improves **resource efficiency without compromising delivery performance**.

- **Proof of Concept:**

The project includes a **software-based simulation that demonstrates the concept of self-organizing logistics systems**.

Instead of relying on external hardware such as IoT sensors or GPS tracking devices, the system operates using:

- a simulated logistics network
- synthetic delivery datasets
- algorithm-driven route optimization

The simulation shows how vehicles can dynamically adjust routes and merge delivery paths using algorithmic decision-making. This serves as a **proof of concept for intelligent logistics coordination**, where optimal routing and resource utilization emerge automatically from the optimization algorithm.

12. CONCLUSION (PHASE 1)

The proposed **Volumetric Fusion A*** approach provides a structured mathematical framework for improving resource utilization in logistics networks. By extending the principles of the A* search algorithm and integrating recent research developments from 2024, the system introduces a novel **Fusion heuristic** that allows delivery routes to merge at strategic nodes. This mechanism enables vehicles to share delivery paths and balance cargo loads more effectively.

A major objective of this approach is to address the **“Empty Miles” problem**, where vehicles travel partially loaded or completely empty between delivery points. By detecting opportunities for route consolidation and load balancing, the proposed model reduces unnecessary travel while increasing volumetric utilization of vehicles.

The framework is designed to be **scalable**, meaning it can be applied to both small distribution networks and large logistics systems with multiple vehicles and delivery nodes. The graph-based modeling of the logistics network allows the algorithm to evaluate multiple routing possibilities efficiently and select routes that maximize utilization while minimizing travel cost.

Future Work

In the next phase of this project, the conceptual framework will be implemented as a **Python-based simulation system**. This implementation will include:

- Integration of **real-world geographic coordinates** for logistics nodes
- Construction of a road network graph representing delivery routes
- Implementation of the Volumetric Fusion A* algorithm for multi-vehicle routing
- Visualization of optimized routes using map-based tools